pfi

Python for Fixed Income

COURSE 10 · IMPLEMENTING RAG

Index Refresh and Corpus Drift

WEEK 3 · DAY 3

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Today: an index that follows its corpus

A 10-K index and a new 10-Q

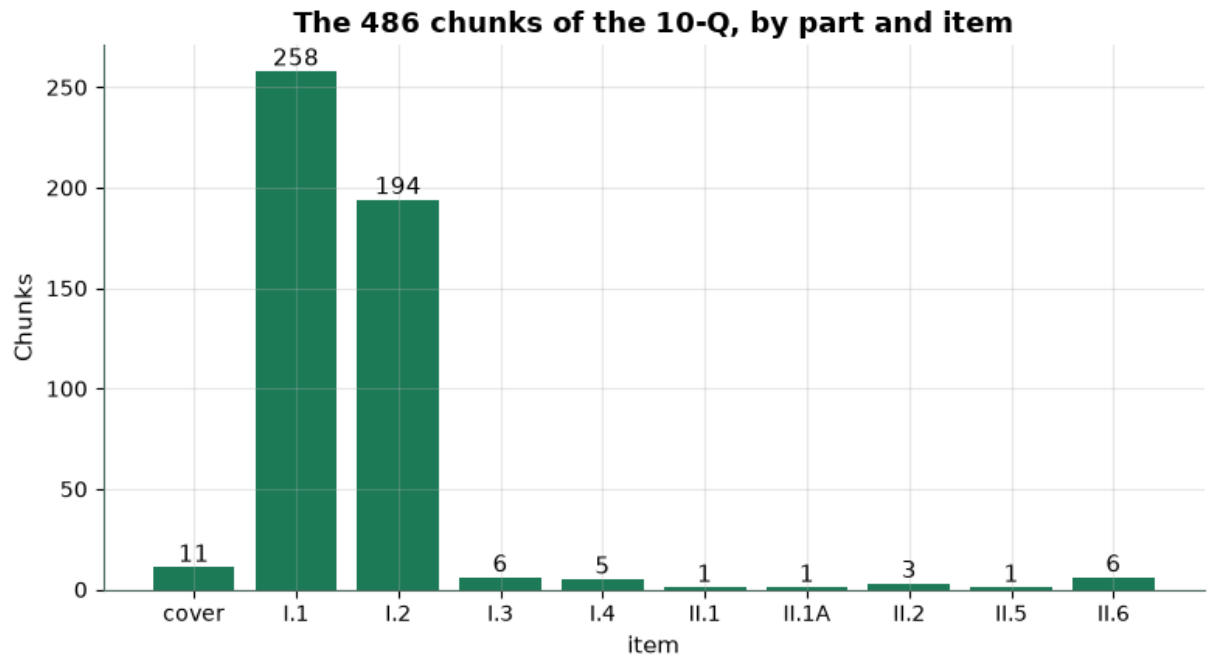
Index	1,158 chunks, MiniLM, 512/64
New filing	10-Q, 486 chunks (Q2 2026)
Corpus	1,644 chunks
Saved answers	9, replayed; no key
Torch	MiniLM, cross-encoder, CPU

Today

1. A corpus changes
2. A chunk's identity: content_hash
3. Refreshing the index
4. As-of dates
5. Three questions, three dates
6. A stale cache
7. A new model means embedding everything
8. Describing the drift

Embed only what is new; name a chunk by its text; say which date a figure is as of; describe drift, never explain it.

A new filing, an amended one, a withdrawn one



Three changes, three jobs

a new filing

new chunks appear; the old ones are untouched. Index: embed the new chunks only.

an amended filing

some chunks keep their text; some change; the ids of the chunks after the change move. Index: embed the changed chunks only, and drop the old ones.

a withdrawn filing

its chunks disappear; the other filings' chunks are untouched. Index: drop its rows; embed nothing.

1,158 + 486 = 1,644 chunks. The first 1,158 are the old index's, unchanged. I.1 is Part I, Item 1.

How it works: the content hash

The need	A name for a chunk that does not change when the chunk moves; an id carries a running number, so an insertion changes every id after it.
The hash	normalize_text first (look-alike characters made plain, white space squeezed), then the SHA-256 of what is left: 64 hexadecimal characters.
Same, other	The same text gives one hash wherever it sits; one changed character gives another; doubled spaces give the same.
You control	Nothing but the text: the name depends on the content alone.
What comes out	The key every refresh matches on, the fingerprint a manifest records, the stamp a saved answer keeps (13.6).

The content hash names a text, not a place

One chunk, two edited copies

the chunk as chunked	`d7ae136072f5` **yes**
the same text with doubled spaces and a blank line at the end	`d7ae136072f5` **yes**
a copy with one number changed (160,700 to 160,800)	`112c1d077216` **no**

7 texts appear twice in the 10-K

```
e7aebf01af11  7-027-0311 and 8-058-0631

9fa8c7ac28ea  7-027-0312 and 8-058-0632

c0ccf571f653  7-038-0427 and 8-076-0829

adf5e50bef29  7-038-0428 and 8-076-0830
```

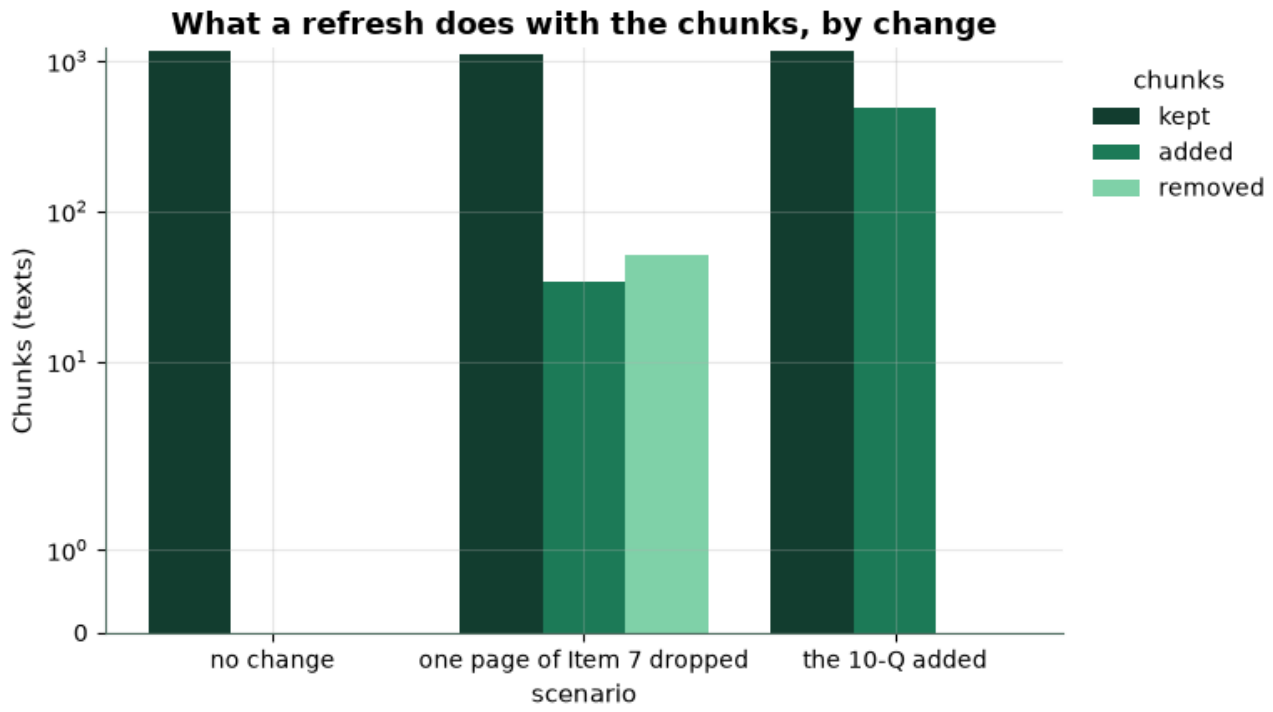
1,158 chunks, 1,151 hashes. Item 7 and a note in Item 8 print the same covenant paragraph.

normalize_text first: look-alike characters made plain, white space squeezed. A chunk id carries a running number, so it is not an identity.

How it works: refreshing the index

Stale	An index goes stale the day its corpus moves; a rebuild embeds every chunk again when most have not changed.
Inputs	The index as it stands (chunks, matrix), the chunk table now, a function that embeds a list of texts, the old manifest, the model key.
Match by hash	A known hash copies its old row; the unknown ones are embedded once, as one list; an old text that no longer occurs is dropped.
Refuse a mix	A model that is not the manifest's raises: rows from two models are not one search space (13.7).
What comes out	The new matrix in the new table's order, the counts kept, added, and removed, and the manifest diff.

Kept, added, removed: only the new texts are embedded



kept / added / removed / embedded

no change

`1,158 / 0 / 0 / 0`

one page of Item 7 dropped

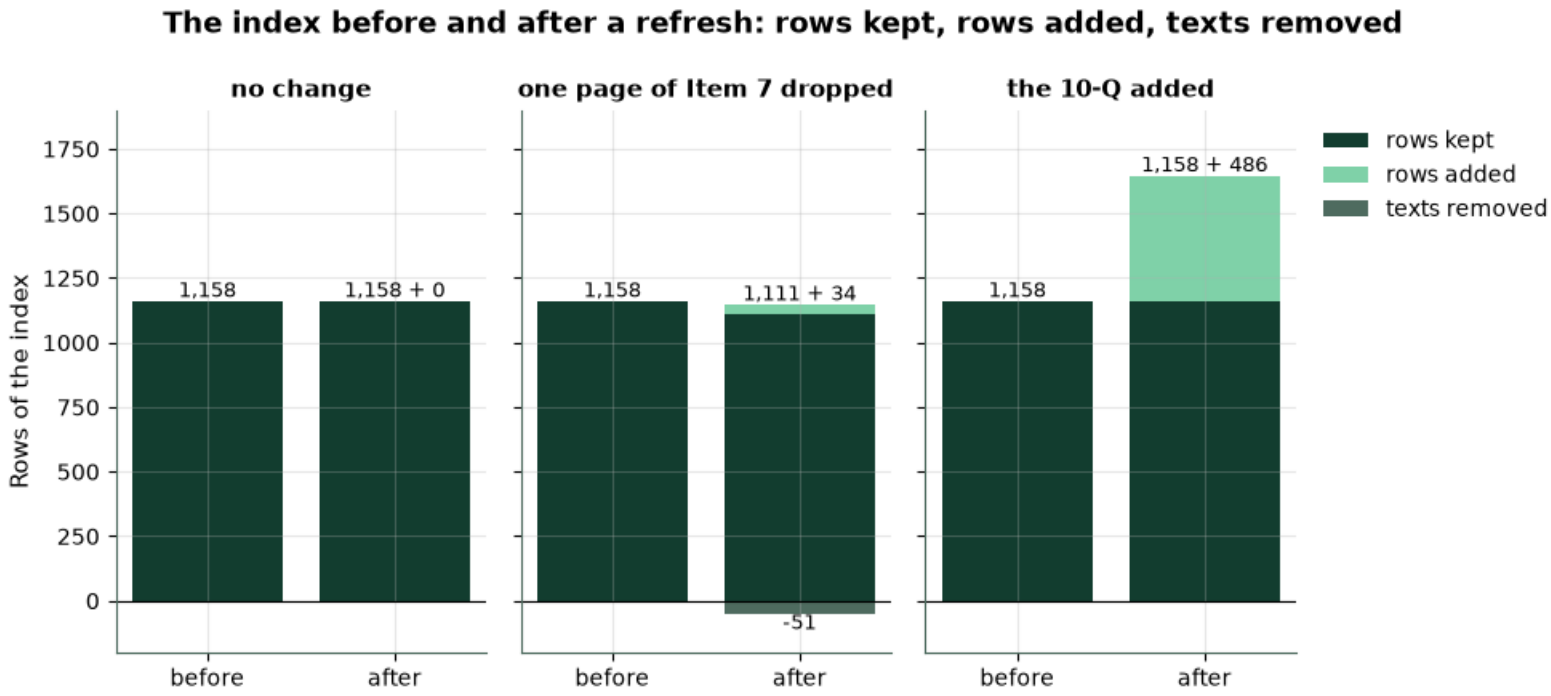
`1,111 / 34 / 51 / 34`

the 10-Q added

`1,158 / 486 / 0 / 486`

The 10-Q: kept 1,158, added 486, removed 0, 486 texts embedded. A model that is not the manifest's raises.

The index before and after each refresh



Rows after each refresh: no change 1,158 (1,158 kept + 0 added); one page of Item 7 dropped 1,145 (1,111 kept + 34 added); the 10-Q added 1,644 (1,158 kept + 486 added). The light segment is the bill, the only part a refresh pays a model for; the texts removed hang below the axis.

Ids move, texts do not

After one page of Item 7 is dropped

699

chunks hold a text the old index holds, under another id

85

ids now hold other text

The manifest diff for the 10-Q

```
chunks: the count

matrix_sha256: the matrix

documents.boyd10q: the new document

not documents.boyd10k: unchanged
```

Excel side by side (tested in Excel)

`=SUMPRODUCT(--EXACT(old,A2))>0`

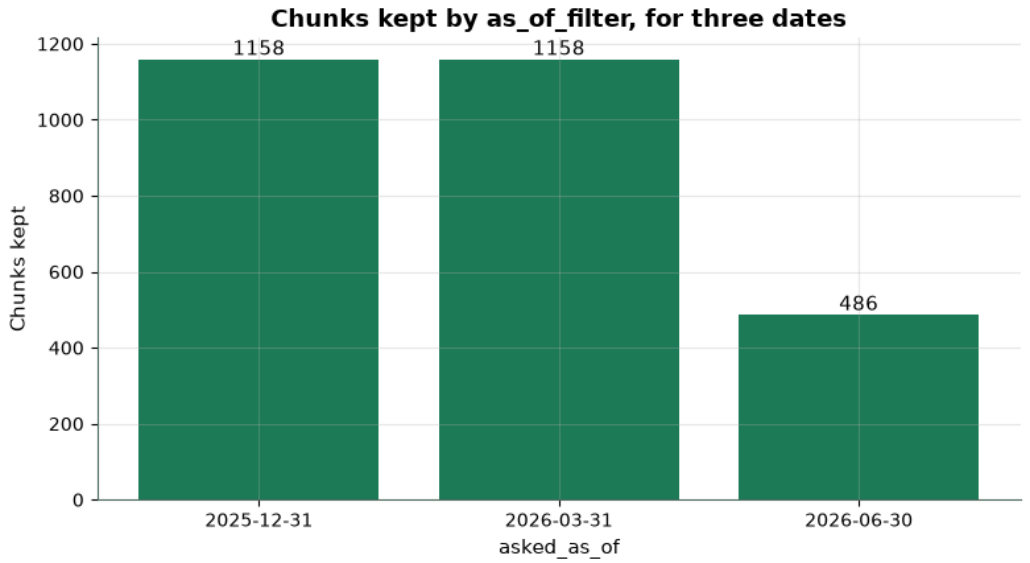
212 of 254 rows TRUE, equal to Python

A refresh keyed by id or place returns text for every search and shows each text with another chunk's score.

How it works: as-of dates

Two filings	One issuer, two filings, different figures for the same thing, each true at its own date; a question with no date is ambiguous.
The mask	Every chunk carries as_of (corpus_chunks); as_of_filter keeps the latest dated document on or before the date asked, and every undated one.
No clock	The date is an argument: the same question and date give the same rows today and next year; a date before every filing raises.
You control	The date, and through corpus_chunks the date each document carries.
What comes out	A boolean mask the length of the chunk table, which a retriever's rows, embeddings, and BM25 index are cut by (13.5).

The latest filing on or before the date asked



asked -> kept (dated)

```
2025-12-31  boyd10k  (2025-12-31)

2026-03-31  boyd10k  (2025-12-31)

2026-06-30  boyd10q  (2026-06-30)
```

ValueError: no dated document is as of 2025-06-30 or earlier

Excel side by side (tested in Excel)

```
=MAXIFS($A$2:$A$3,$A$2:$A$3,"<="&D2)
```

0 when there is no filing: 0, no error

The filter returns the latest filing on or before the date; the figures in it stand at that filing's date. The date is an argument, never the clock.

Every setting of today's functions

Function	Setting	Default	What it controls	Today passes
content_hash	text	required	the chunk's text; normalize_text runs first, so spacing and look-alike characters do not count	each chunk's text
refresh_index	old_chunks	required	the chunk table of the index as it stands	old_chunks (the 10-K's 1,158)
refresh_index	old_matrix	required	its embeddings, one row per old chunk	old_matrix (day 4's committed rows)
refresh_index	new_chunks	required	the chunk table as the corpus is now	corpus (both filings), amended, or old_chunks again
refresh_index	embed_fn	required	a function from a list of texts to their rows; run once, on the new texts only	embed_new (MiniLM, counting what it is given)
refresh_index	old_manifest	required	the record of what the old index was made from; its model is checked	old_manifest
refresh_index	model_key	required	the model the new rows are embedded with; must be the manifest's, or ValueError	"minilm" ("bge_small" in 13.7, refused)
as_of_filter	chunks	required	a chunk table with an as_of column (corpus_chunks wrote it)	corpus
as_of_filter	as_of	required	the date asked, ISO text; the latest dated document on or before it is kept, undated ones always	2025-12-31, 2026-03-31, 2026-06-30

17 settings in the notebook's table across five functions, 16 required; the 8 of index_manifest and corpus_chunks are read there. A desk chooses two: model_key and the date.

Which filing's figure does each answer state?

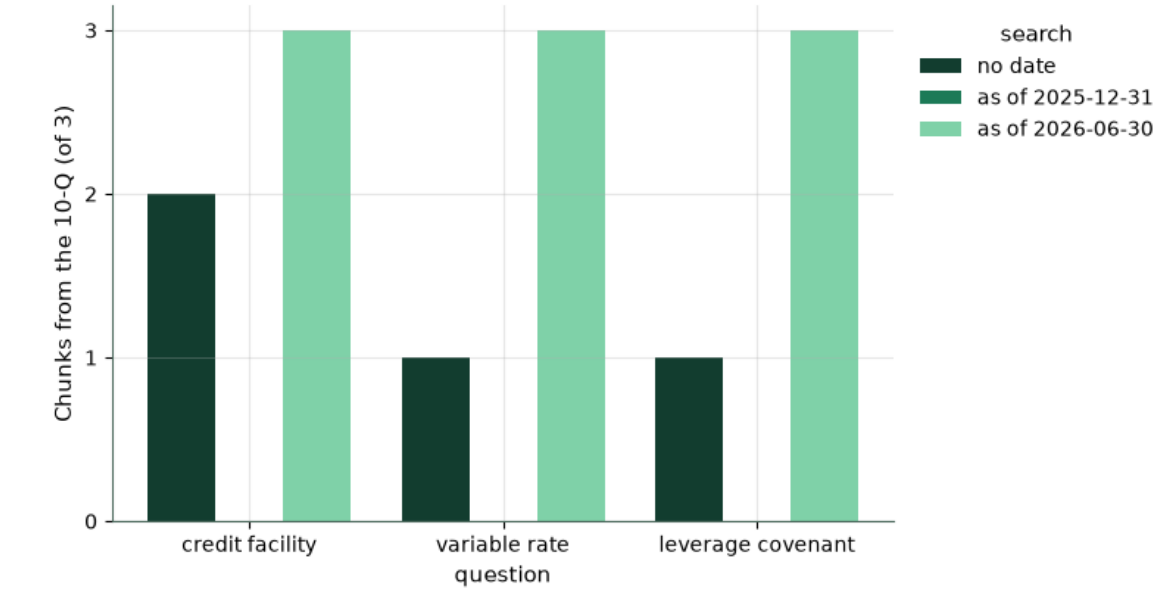


Figure each answer states

credit	no date	neither
credit	2025-12-31	2025-12-31
credit	2026-06-30	neither
variable	no date	both
variable	2025-12-31	2025-12-31
variable	2026-06-30	2026-06-30
leverage	no date	both
leverage	2025-12-31	both
leverage	2026-06-30	both

`chunker=recursive 512/64, embedder=minilm@1110a24, index=faiss exact, retrieve=hybrid k=3, rerank=cross-encoder, expand=0, model=gpt-4.1-mini, temperature=0, top_p=1, max_tokens=300`

With no date the search takes whichever chunks rank first. The as-of scope is the date in the search; the question asks the answer to say it.

The chunk that stopped short

[boyd10q; Section I.2; c0417], last lines

```
June 30, December 31,

(In millions) 2026 2025

Revolving Credit Facility $ 315.0 $ 135.0

Term A Loans
```

(the chunk ends here)

As of 2026-06-30

answer: $715.0 million

key: $741.1 million

The Swing Loan line and the total row are in the next chunk, c0418, which was not one of the three. The model added the two lines it had.

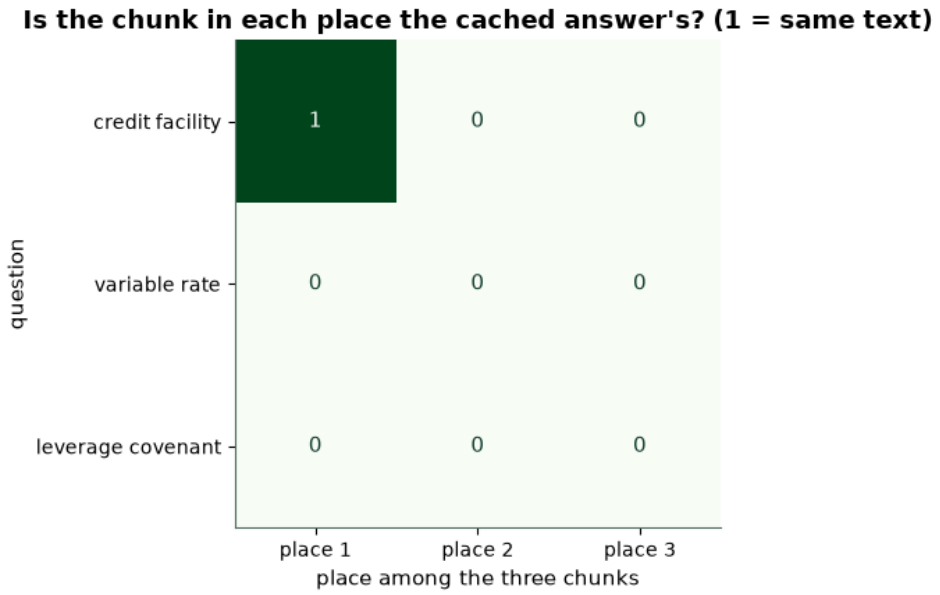
Source: Boyd Gaming Corporation, Form 10-Q for the quarter ended June 30, 2026, filed 2026-07-30, accession 0001437749-26-025062, Part I, Item 2.

A retrieval miss by the figure rule: the key's figure was not in the chunks sent.

How it works: a stale cache

Saved answers	A service keeps its answers; each is true of the chunks it was written from, and a refresh can move those chunks.
Keep hashes	Beside each saved answer, the content hash of each chunk behind it, in order.
Compare	Ask again, hash the chunks that come now, compare the two lists: equal means the same text; different means the index moved under it.
Of the text	Hashes of the text, never of the ids (13.2); a changed set is a reason to ask again, not yet a wrong answer.
What comes out	One boolean per saved answer, stale or not, and the count of places where the chunk is still the same.

Saved answers, checked by the hashes of their chunks



question / in place / among three / stale / fresh answer states

credit facility

`1 of 3 / 1 of 3 / yes / neither`

variable rate

`0 of 3 / 2 of 3 / yes / both`

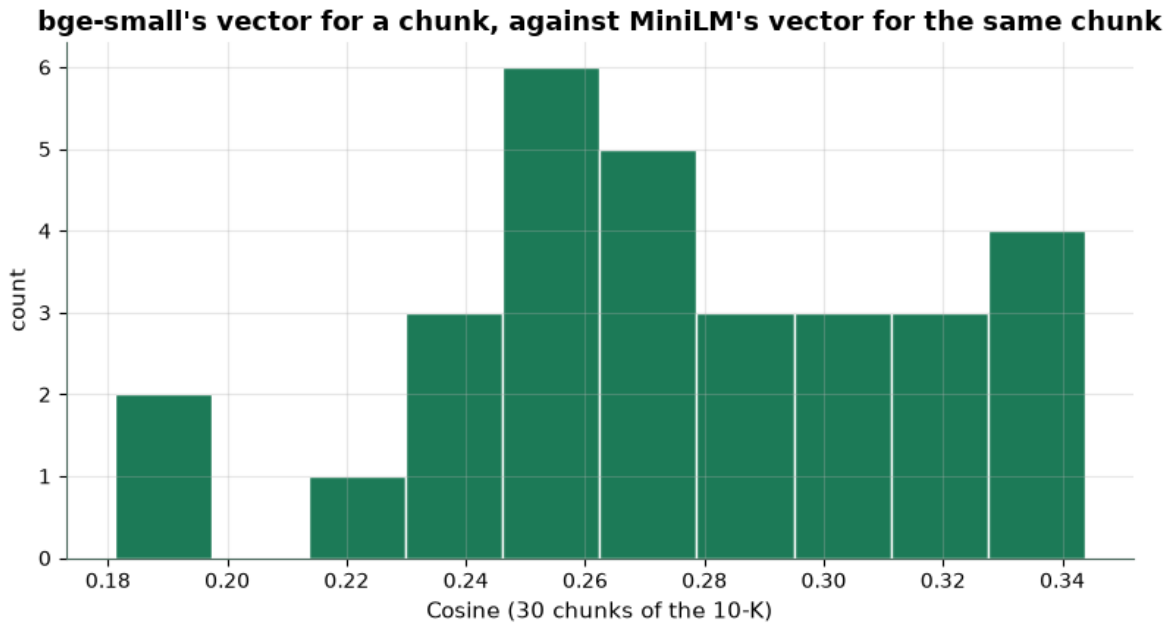
leverage covenant

`0 of 3 / 2 of 3 / yes / both`

Stale before: credit facility no, variable rate no, leverage covenant no. After: credit facility yes, variable rate yes, leverage covenant yes.

The test says the chunks differ, nothing about the figure. A changed set is a reason to ask again.

Same width, another space



Cosine of two vectors for the same chunk

MiniLM live with MiniLM committed

```
min 1.0000  mean 1.0000  max 1.0000
```

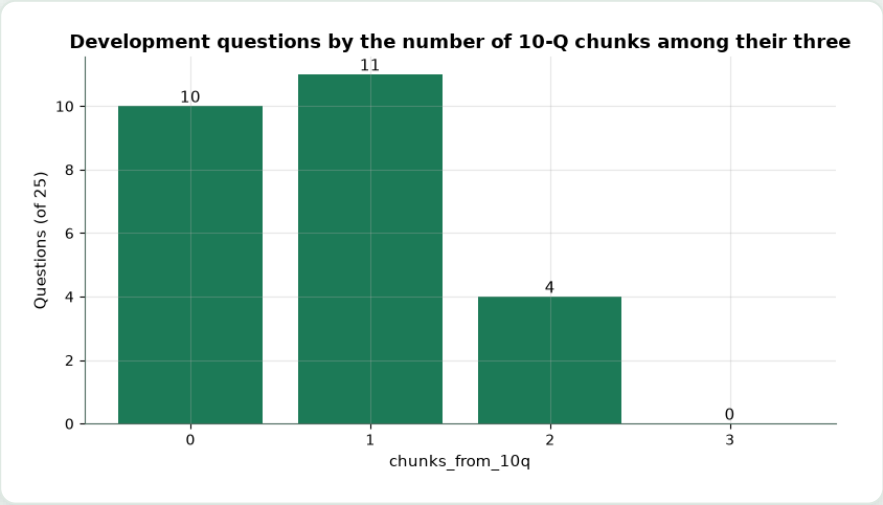
bge-small live with MiniLM committed

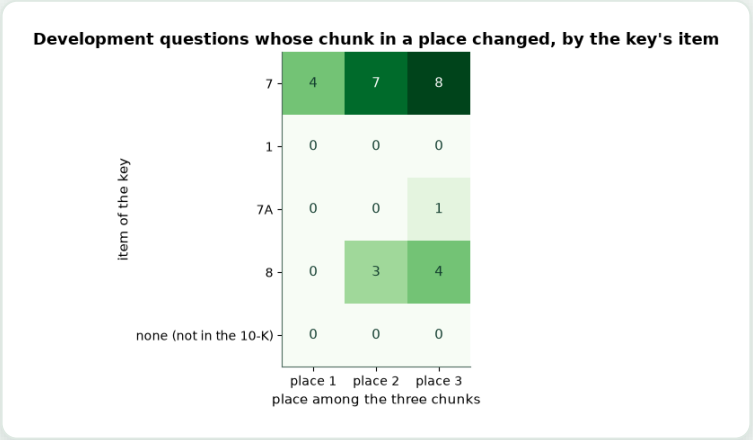
```
min 0.1811  mean 0.2764  max 0.3438
```

```
refresh_index: the index was built with minilm, not bge_small: re-
embed every chunk with one model
```

A row from one model in another's matrix is a vector in the wrong space, and a search compares it without any error.

What moved when the 10-Q arrived





25 development questions

29.6%
of the corpus is new

15 of 25
have other chunks

4
have another first chunk

`First chunk changed: dev04, dev07, dev08, dev14`

A description, and it stops there: the numbers say which questions moved, not why any figure differs between the filings.

Excel side by side, and Git

Excel side by side (tested in Excel)

`=SUMPRODUCT(--EXACT(old,A2))>0`	212 of 254 TRUE, equal to Python
`=COUNTIF(old,A2)`	#VALUE! on 253 of 254 chunk texts; 255 characters found, 256 not
`=MATCH(A2,old,0)`	#VALUE! on 253 of 254
`=MAXIFS($A$2:$A$3,$A$2:$A$3,"<="&D2)`	46022 for 2025-12-31 and 2026-03-31, 46203 for 2026-06-30
`A date before every filing`	MAXIFS returns 0, no error; as_of_filter raises
`pd.ExcelWriter workbook`	chunk ids and hashes read back as text

Git: index_manifest.json before and after the refresh; 61 tests.

The hash column is computed in Python; no hash was reproduced in Excel.

The AI check

The docstring is the prompt. The retrieval code that comes back runs and looks finished.

1

Docstring

Units, conventions, and what the code must never let through: what you asked for.

2

Code back

Written for the course in the style of an AI answer. Read it.

3

Test first

Plant the case the docstring promises to catch. Run it: it fails.

4

Fix

Find the rule it broke, fix it, run the test again.

Today: the index keeps old rows by position, so a chunk put in front gives every old row to the chunk above it.

Code that runs is not the same as code that is right.

What you can do now

- content_hash names a text, not a place
- refresh_index: keep known hashes, embed the rest, refuse a model mix; draw before and after
- as_of_filter: the latest filing on or before the date
- A changed set of chunks is a reason to ask again
- Describe drift; never explain it

Week 3, Day 4 Serving the retriever and what it costs